14. Exploring Pro-Code Extension of a Zoho Creator Application Using Zoho Catalyst / Custom Functions

Aim

To explore the extension of the Student Admission Enquiry Management application in Zoho Creator using custom Deluge functions and Zoho Catalyst serverless capabilities for reusable calculations, secure API integrations and advanced admission-data processing.

Theory

Zoho Creator provides forms, reports, workflows and Deluge scripting for rapid development. Admission management may also need specialized processing beyond standard low-code components, such as generating admission acknowledgements, verifying supporting documents and integrating with external systems.

- Custom Deluge Functions: Reusable user-defined functions can perform validations, fee calculations and event-driven operations within the Admission_Enquiry_Form and its workflows.
- Zoho Catalyst Serverless Integration: A separate backend, written in a supported programming language, can handle complex file processing, document generation or API-based verification. Creator can exchange data with that backend through an authenticated connection.
- Security and reliability: Sensitive applicant information should be transmitted only through approved endpoints with proper authentication, least-privilege access, validation and logging.

Procedure

Step 1: Creating a Custom Deluge Function in Zoho Creator

- Open the Student Admission Enquiry Management application in Zoho Creator and navigate to the functions or workflow configuration area.
- Create a reusable function, for example Calculate_Admission_Fee, accepting the application fee and scholarship percentage as inputs.
- Implement and test the calculation, then call the function from a suitable Admission_Enquiry_Form workflow.

Illustrative Deluge logic (adapt field/API syntax when configuring):

```
fee = 750.00;
scholarshipPercent = 25.00;
netFee = fee - (fee * scholarshipPercent / 100);
info netFee;
```

For this illustrative input, the calculated net fee is 562.50. The calculation is an example, not a confirmed rule in the running application.

Step 2: Setting Up Zoho Catalyst

- Sign in to Zoho Catalyst and create a project for Student Admission Enquiry Management.
- Define a serverless function in an available supported runtime to process a selected task, such as producing an admission acknowledgement PDF or validating uploaded document metadata.
- Test the function independently using non-sensitive sample admission records before allowing real applicant data.

Step 3: Connecting Zoho Creator to Zoho Catalyst

- Deploy the serverless function and obtain the appropriate authenticated endpoint for the project.
- Create an authorized connection in Zoho Creator, then invoke the function with limited necessary fields such as an enquiry record identifier.

Illustrative Deluge invocation (placeholder values; not a deployed URL):

```
response = invokeurl
[
    url : "https://YOUR_AUTHORIZED_ENDPOINT"
    type : POST
    parameters : {"enquiryId": input.ID}
    connection : "YOUR_CONNECTION_NAME"
];
info response;
```

Step 4: Testing and Validating the End-to-End Workflow

- Submit a test record through Admission_Enquiry_Form and verify that the corresponding form workflow is triggered.
- Inspect Creator and Catalyst execution logs for request success, response contents and error handling.
- Check that the output is linked to the correct enquiry record in Admission_Enquiry_Report and that applicant information is access-controlled.

Result

The proposed pro-code extension of the Student Admission Enquiry Management application was documented using custom Deluge functions and Zoho Catalyst. The approach illustrates how reusable admission-fee logic and external serverless processing could extend Zoho Creator. Deployment, end-to-end API connectivity and successful execution require implementation and testing; no verified Catalyst execution screenshots were supplied for this experiment.